Higher-order Markov Models can Help

- A bigram model will probably assign a similar probability to the sentences:

The birds I saw last night **sing** beautifully

The birds I saw last night **sings** beautifully

Or:

He saw himself in the mirror

He saw herself in the mirror

Higher-order Markov Models can Help

198015222 the first
194623024 the same
168504105 the following
158562063 the world
…
14112454 the door

23135851162 the *

197302 close the window
191125 close the door
152500 close the gap
116451 close the thread
87298 close the deal

3785230 close the *

3380 please close the door
1601 please close the window
1164 please close the new
1159 please close the gate
…
0 please close the first

13951 please close the *

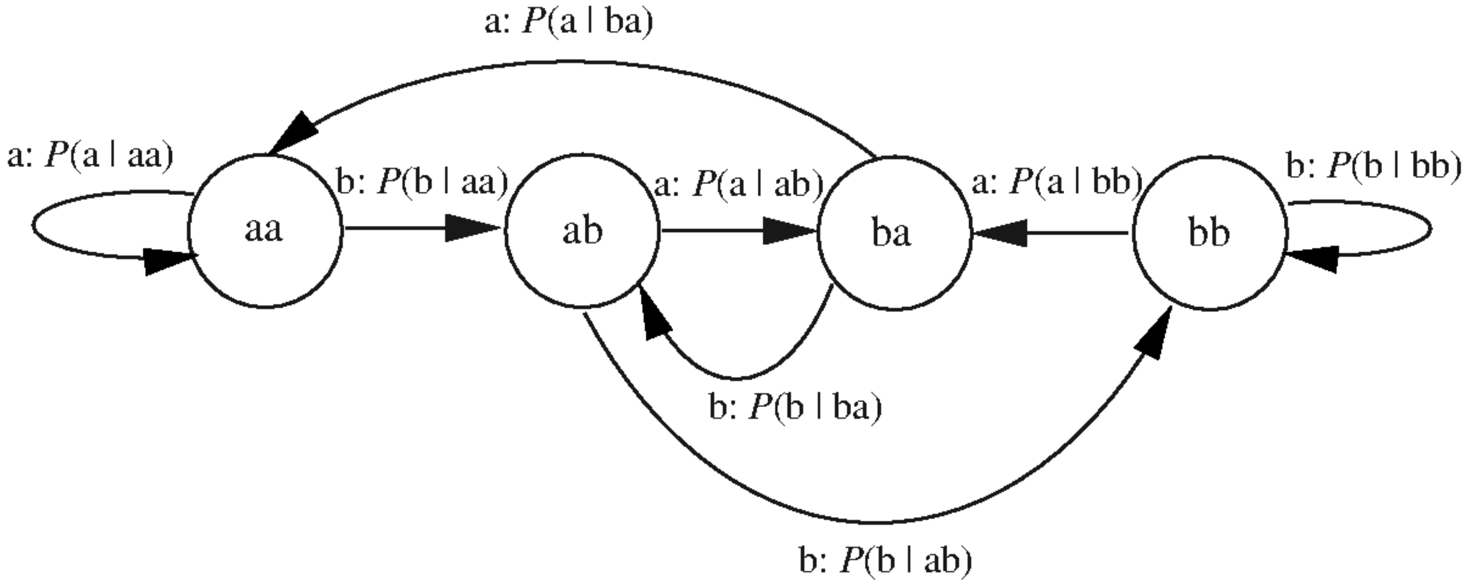
Higher-order Models

- k-th order Markov model

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i | x_{i-1}, \ldots, x_{i-k})$$

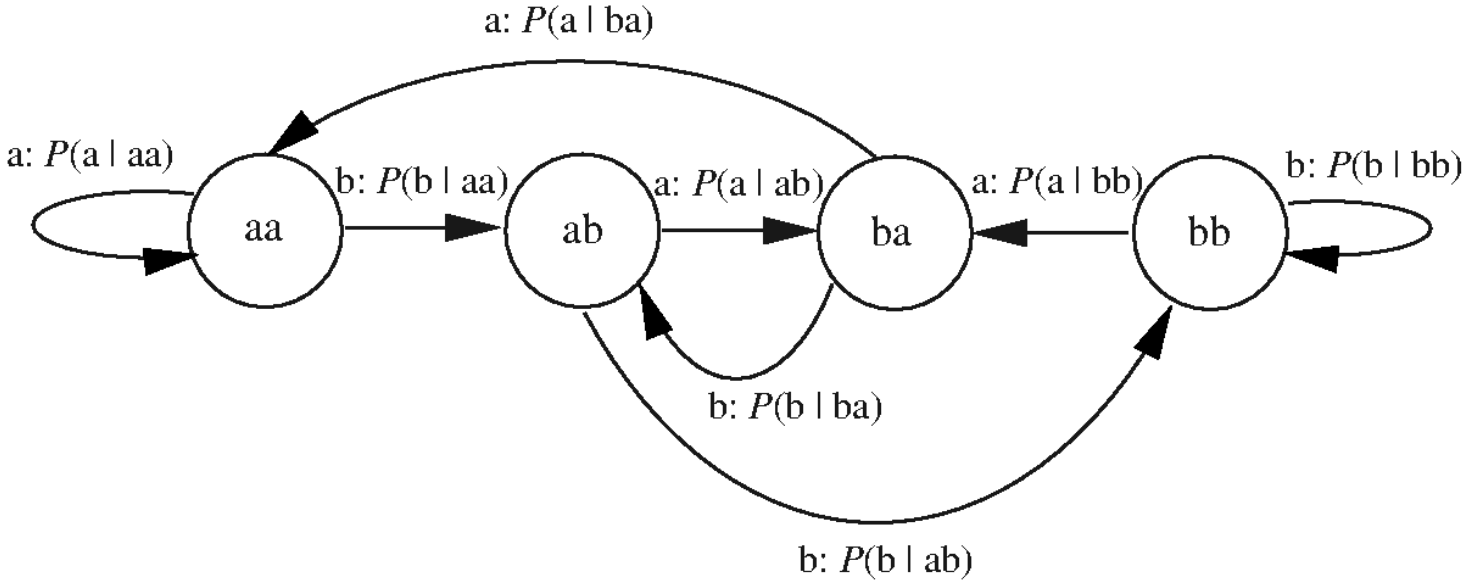
Higher-order Markov Language Models

- k-gram models are equivalent to bi-gram models where the state space (the words) are k-tuples of the bi-gram state space
- So we think of these models as modeling the transition between a k-tuple of states to a k-tuple of states, where the first *k-1* coordinates of x_n must be equal to last *k-1* coordinates of x_{n-1}



Higher-order Markov Language Models

- The difficulty: the number of parameters increases exponentially with *k*.
- So a k-gram language model would have about $|\text{Words}|^k$ entries in its transition matrix



Learning Higher-order Language Models

- A maximum likelihood estimator for a k-th order language model is:

$$q_{ML}(x_m = w_j | x_{m-1} = w_{i1}, x_{m-2} = w_{i2}, \ldots, x_{m-k} = w_{ik}) = \frac{count(w_{ik}, \ldots, w_{i1}, w_j)}{\sum_{j'} count(w_{ik}, \ldots, w_{i1}, w_{j'})}$$